

Code-Layout, Readability and Reusability

Date	03 October 2026
Team ID	17
Project Name	Fitbuddy-AI Fitness plan Generator
Department	Computer Technology
Institution / University	Madras Institute Of Technology / Anna University
Academic Year / Semester / Section	3 / 5 / Batch A
Faculty Mentor / Guide	Jayachitra V P
Team Leader	Bharath Jeyakkumar S
Sprint 1 Start / End	28-09-2026 / 28-09-2026
Sprint 2 Start / End	29-09-2026 / 30-09-2026
Sprint 3 Start / End	01-10-2026 / 02-10-2026
Team Roles	Abdul Wahith M — Backend & AI Integration; Kamalesh T — UI / Frontend Development; Bilu Besto H — Validation, Feedback & Reliability; Bharath Jeyakkumar S — Team Leader / Testing & Quality Assurance; Prasannavelan K — Database, Documentation & Demo Support
Maximum Marks	2 Marks

Repository Layout

FitBuddy Project Code Layout

```
fitbuddy/
├─ app/
│   ├── main.py
│   ├── routes.py
│   ├── config.py
│   ├── schemas.py
│   ├── database.py
│   ├── gemini_client.py
│   ├── gemini_generator.py
│   ├── gemini_flash_generator.py
│   ├── updated_plan.py
│   ├── mock_ai.py
│   ├── templates/
│   └─ static/
├─ tests/
├─ scripts/list_models.py
├─ tools/
├─ docs-src/
└─ requirements.txt
```

FitBuddy Project Code Layout

Readability

- Responsibilities are separated into configuration, schemas, database, routes and AI modules.
- Shared plan-generation logic is used by both HTML and JSON API routes.
- Validation errors are converted into field-oriented messages.
- Environment configuration avoids hard-coded credentials.

Reusability

- `gemini_client.py` centralizes model creation, retries and fallbacks.
- Generator modules can be replaced without rewriting route handling.
- JSON API endpoints allow a future independent frontend.
- Mock AI makes the application testable without external AI calls.